Robert M. Gower

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modern optimization | adaptive methods | stochastic optimization | machine learning | randomized numerical linear algebra

Profile

I work on the mathematics of modern optimization: why methods such as Adam, Muon, Polyak step-sizes, and non-Euclidean gradient descent work, fail, and sometimes come up with new methods. My current research sits between optimization theory and machine-learning practice, with a focus on simple algorithms that exploit structure. Previously, I worked on randomized numerical linear algebra, automatic differentiation, and quasi-Newton methods.

Appointments and Education

2021–present	**Research Scientist**, Flatiron Institute · Center for Computational Mathematics, New York
2021	**Visiting Scientist**, Google Research · New York
2020	**Visiting Scientist**, Facebook Research · New York
2017–2021	**Assistant Professor**, Telecom Paris, Institut Polytechnique de Paris · Machine learning and optimization
2016–2017	**Laureate Fellow**, Fondation Sciences Mathematiques de Paris · ENS and Inria, Paris
2012–2016	**Ph.D.**, University of Edinburgh · Optimization and Operations Research
2011–2012	**Analyst in Risk and Validation**, Itaú Unibanco · São Paulo
2011	**M.Sc.**, University of Campinas · Applied Mathematics

Recent Highlights

2026	*The Polar Express* was an **oral** at ICLR and received the Outstanding Paper Honorable Mention.
2026	*The Polar Express* is part of the record-holding entries in the modded-nanogpt and nanochat LLM speedrun competitions (checked October 1, 2026).
2026	*Non-Euclidean Gradient Descent Operates at the Edge of Stability* was an **oral** at ICML.
2025	*In Search of Adam's Secret Sauce* was an **oral** at NeurIPS.

Selected Papers

- N. Amsel, D. Persson, C. Musco, **R. M. Gower**. *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm*. ICLR 2026, oral; Outstanding Paper Honorable Mention.
- A. Orvieto, **R. M. Gower**. *In Search of Adam's Secret Sauce*. NeurIPS 2025, oral.
- R. Islamov, M. Crawshaw, J. Cohen, **R. M. Gower**. *Non-Euclidean Gradient Descent Operates at the Edge of Stability*. ICML 2026, oral.
- D. Cai, C. Modi, L. Pillaud-Vivien, C. Margossian, **R. M. Gower**, D. M. Blei, L. K. Saul. *Batch and Match: Black-Box Variational Inference with a Score-Based Divergence*. ICML 2024, spotlight.
- **R. M. Gower**, P. Richtarik. *Randomized Iterative Methods for Linear Systems*. SIAM J. Matrix Anal. Appl., 2015; Leslie Fox Prize, 2nd place.

Honors, Service, and Mentoring

- **Prizes**: ICLR Outstanding Paper Honorable Mention (2026); IMA Leslie Fox Prize in Numerical Analysis, 2nd place (2017); FSMP Laureate Fellowship (2016).
- **Reviewing rewards**: ICML Gold Reviewer (2026); NeurIPS Top Reviewer (2025); ICML Best Reviewer, top 10% (2021); NeurIPS Top 10% Reviewer (2020); ICML Top 5% Reviewer (2019).
- Area chair for NeurIPS and TMLR; reviewer for ICML, JMLR, SISC, SIOPT, Mathematical Programming, and related venues.
- **Mentored** Ph.D. students, interns and postdocs including Nidham Gazagnadou, Rui Yuan, Samy Jelassi, Si Yi Meng, Fabian Schaipp, Slavomir Hanzely, Aaron Mishkin, David Persson, Michael Crawshaw, and Tetiana Parshakova.